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On Bell's inequality and entanglement

Quantum Physics

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1 Introduction

Quantum mechanics is a fundamental theory in modern physics, capable of describing phenomena in a precise scale. Even though it is described probabilistically, the quantum theory and the wavefunction state description are considered complete, enough and able to build our real world. Historically, the completeness of this theory had been a debate for a long time, carried by the EPR paradox (A. Einstein, B. Podolsky & N. Rosen, 1935) and the response to it (N. Bohr, 1935). All this led to further experimentation and proofs (J. S. Bell, 1964), which mathematically opposed with the EPR paradox.

The objective of this work is to analyze the historical evolution of this debate, from the EPR to the Bell inequality and the subsequent experimental verification. We will start with the historical context, introducing technical concepts and the previous theory before the 1935 EPR paper, then deriving the Bell inequality and finally providing some mathematical simulation of experiments related with the debate.

1.1 Historical context

1.1.1 Schrödinger equation

All this extended discussion had origin in 1926, when Austrian physicist Erwin Schrödinger formally exposed the Schrödinger equation in a German journal titled *Annalen der Physik*. This is the fundamental equation of non-relativistic quantum mechanics, and the one for which we provide a simple analysis to introduce further definitions, calculations and proofs.

The general Schrödinger equation for a single particle in 3D is expressed as¹:

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r},t) + V(\mathbf{r},t)\Psi(\mathbf{r},t) = i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) \quad (1)$$

Solving the Schrödinger equation allows us to determine the different quantum states in which a particle can be described, that is, the wavefunctions containing information about its measurable physical quantities, such as momentum, position, and angular momentum. There are two different approaches to interpret this equation, the **TISE**² for a single particle in 3D:

$$\hat{H}\psi(\mathbf{r}) = E\psi(\mathbf{r}) \quad (2)$$

and the **TDSE**³ for a single particle in 3D:

$$\hat{H}|\Psi(\mathbf{r},t)\rangle = i\hbar\frac{\partial}{\partial t}|\Psi(\mathbf{r},t)\rangle \quad (3)$$

where $\hat{H}$ is the Hamiltonian Operator and E the energy.

Solving the TISE can primarily determine quantized energy levels, where the wavefunctions (eigenfunctions in this case) correspond to stationary states. It simply proves the existence of such allowed energy levels, explaining, for instance, the reason why electrons in atoms can only exist in specific orbitals with their associated specific energy values (eigenvalues of each eigenfunction). We are not delving into this equation due to its lack of direct alignment with our project.

Solving the TDSE can principally determine the evolution of the quantum state in general. As an application it can either determine the probability distribution of the particle having certain properties using the **Born Rule** or calculate expectation values for any measurable physical quantity.

1.1.2 Quantum Entanglement

A key concept can be deduced from the TDSE. Considering an interaction of two particles of position $\mathbf{r}_1$ and $\mathbf{r}_2$, we write the general form as:

$$\hat{H}|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle = i\hbar\frac{\partial}{\partial t}|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle \quad (4)$$

¹ $\hbar$ =Dirac Constant= $\frac{h}{2\pi} = 1.055 \cdot 10^{-34} J \cdot s$, m =Mass, ∇^2 =Laplacian Operator, Ψ =Wavefunction, V =Potential

²Time-Independent Schrödinger Equation

³Time-Dependent Schrödinger Equation

Developing and solving this equation, considering a combined quantum state $|\Psi\rangle$ as the solution, we can then differentiate two cases:

1. **Separable solution:**

It means that the solution obtained solving the TDSE is a combined quantum state $|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle$ that can be expressed as a tensor product of two wavefunctions $|\psi(\mathbf{r}, t)\rangle$, each depending on different position variables $\mathbf{r}_1$ and $\mathbf{r}_2$:

$$|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle = |\psi_1(\mathbf{r}_1, t)\rangle \otimes |\psi_2(\mathbf{r}_2, t)\rangle \quad (5)$$

2. **Inseparable solution:**

In this case the solution obtained solving the TDSE is, as well, a combined quantum state $|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle$. However, in this occasion, this combined state cannot be expressed as a tensor product of two wavefunctions depending on different position variables:

$$|\Psi(\mathbf{r}_1, \mathbf{r}_2, t)\rangle \neq |\psi_1(\mathbf{r}_1, t)\rangle \otimes |\psi_2(\mathbf{r}_2, t)\rangle \quad (6)$$

This state with inseparable variables defines mathematically the concept we are introducing: **Quantum Entanglement**.

Fundamentally, this phenomenon describes a pair or a group of particles whose quantum states cannot be described independently of the states of the others. In other words, the information of the whole system is less uncertain than each of its individual parts, a fact that is impossible in classical physics [1].

According to Quantum Mechanics, particles cannot have definite properties like **momentum** or **spin** at any given point. It is upon measurement that their wavefunction collapses into a given state. However, entanglement was found to be such that, if one particle is measured, its entangled particle would too, independently of the distance between them. This led to Einstein describing the phenomenon as "spooky action at a distance" in the 1930s.

To improve the understanding of this phenomenon in our real world, a simple case we can consider is a pair of electrons. Two entangled electrons in a singlet state, described mathematically as:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad (7)$$

will have opposite spin states; either spin up ($\mathbf{S} = +\frac{1}{2}$) or spin down ($\mathbf{S} = -\frac{1}{2}$). But neither of them has a definite spin value until we measure it. That's when the other electron's spin also acquires a definite value.

This is independent of distance, as has been verified experimentally [2]. The moment we observe one electron, we know, for sure, the spin value of the other electron; total spin must always be conserved. This instantaneous correlation was considered impossible at that time due to the fact that no information was able to be transmitted faster than the speed of light⁴. As was later to be found, this effect does not necessarily imply FTL⁵ communication.

1.2 The EPR Paradox

On May 15, 1935, physicists Albert Einstein, Boris Podolsky and Nathan Rosen presented the famous **Einstein Podolsky Rosen Paradox (EPR)** in the American journal *Physical Review*, in a paper titled *Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?* [3]. In this article they manifested the incompleteness of the quantum physics theory, opening a debate of what it actually needed to be considered complete.

They started by making the following assumptions:

1. **Condition of Correctness:**

The consideration of success of a physical theory has a pair of requirements. They are, as revealed in the article, correctness and completeness of the theory. The condition of correctness can be fulfilled if human experiences agree with the conclusion of the theory.

⁴Speed of light: $c \approx 300.000 \frac{km}{s}$

⁵FTL stands for *Faster Than Light*.

2. Condition of Completeness:

The condition of completeness is, as they said, "*every element of the physical reality must have a counterpart in the physical theory*". In other words, every single result in the theoretical description must correspond with an element or phenomenon in the real world.

3. Counterfactual definiteness:

Also called **realism**. This assumption defines a theory that assumes preexisting properties, which can then be uncovered by experiments. This is another way of saying that all the properties of the system are predetermined, therefore allowing for hidden variables to exist. In relation to a very famous philosophical question, it's like saying that a tree falling makes noise whether or not someone hears it.

4. Criterion of reality:

If, without disturbing the system, we can predict with certainty the value of a physical quantity, then there exists an **element of reality** corresponding to that quantity.

5. Principle of locality:

Any system is directly influenced only by its immediate surroundings, so that any second system not having a causal connection with the first cannot affect its qualities. That causal connection is limited by the speed of light, so we are implicitly saying that superluminal communication is not possible.

It is followed by an explanation of an example of the Heisenberg Uncertainty Principle, specifically the case of uncertainty between linear momentum and position of a particle, but we are explaining it without such a complexity. As an example, considering the state defined by the wave function:

$$\psi = e^{(\frac{2\pi i}{h})p_0 x} \quad (8)$$

and the momentum operator:

$$p = (\frac{h}{2\pi i} \frac{\partial}{\partial x}) \quad (9)$$

it can be easily proved that p_0 is the definite eigenvalue of the linear momentum operator for the state ψ :

$$p\psi = (\frac{h}{2\pi i} \frac{\partial \psi}{\partial x}) = \frac{h}{2\pi i} \frac{2\pi i}{h} p_0 \psi = p_0 \psi \quad (10)$$

Thus, as the momentum is fully determined, by calculating the probability of finding the particle between x_1 and x_2 :

$$P(x_1, x_2) = \int_{x_1}^{x_2} \bar{\psi} \psi dx = \int_{x_1}^{x_2} dx = x_2 - x_1 \quad (11)$$

we can describe nothing about the position of the particle, as the probability of finding the particle in such interval is constant and it only depends on the distance of the interval limits.

Following with a more general definition, in a system with two non-commuting operators, each associated with their physical quantity, the measurement of one of them forbids the attainment of any information about the second one without perturbing the system. It is known that both momentum and position exist in real world, so this fact was counterintuitive with the human experiences. This leads to two mutually exclusive solutions, the **incompleteness of the reality description** or the **impossibility of simultaneous reality** of two physical quantities with non-commuting associated operators.

Supposing that the wave function describes completely the physical reality and considering two systems in interaction until a certain time $t_1 > 0$, that is, they no longer influence each other afterwards, it is possible to calculate the combined state via the two-particle interaction Schrödinger equation, even for any $t > t_1$, but then the determination of quantities of each system after the separation would be unable.

For further developments it is necessary to consider two physical quantities with non-commuting associated operators. Each physical quantity has its eigenvalues and their corresponding eigenfunctions, able to define the state of the first system and the corresponding state of the second system, even lacking interaction in that moment. These two physical quantities lead to a pair of wave functions that both belong to the same reality, proving the simultaneous reality of two quantities with non-commuting operators. Hence, the negative supposition of the **incompleteness of the reality description** leads us to falsify the **impossibility of simultaneous reality** as well. Thus, there remains a single possibility with

the previous assumptions: **The quantum-mechanical description of physical reality given by wave functions is incomplete.** [3]

There could still be a misconception, where making the reality simultaneous, in this case, means to measure both quantities precisely in the same moment. Nevertheless, that requires a condition during the measurement, in other words, the reality in this instance depends on the process of measurement taken for the first system, without disturbing the second one, and no reality can be defined this way.

2 Bell's Theorem

2.1 Bell inequalities and proof of the Theorem

In 1964, physicist John Stewart Bell published his famous paper, titled: *On the Einstein Podolsky Rosen Paradox* [4]. In this paper, Bell gave a mathematical proof that hidden variable theories are incompatible with quantum mechanics⁶, and presented a way to verify it experimentally.

He starts by making the same two main assumptions as in the EPR paper, namely **locality** and **counterfactual definiteness**⁷, to later provide an inequality that is incompatible with quantum mechanics, proving that some of the assumptions must be, in that case, incorrect.

The proof that he provided is rather complicated, so we will cover here some simpler versions to make it simpler to understand with our current knowledge and to grasp the basic idea and cleverness of his argument.

2.1.1 A simple proof of Bell's inequality

We introduce here a demonstration, with a little bit of basic quantum mechanics, published by Lorenzo Maccone [5]. We start by assuming **realism** and **counterfactual definiteness**; those are our principal explicit assumptions. Nevertheless, there are some concepts we must assume implicitly for the whole thing to work, the most important one being **freedom of choice** [6], also called **no superdeterminism**. This implies that our decision to make a certain measurement is independent of the experiments we are performing. That is, our choices of what properties to measure are not predetermined. We'll see the importance of this assumption later on.

Now with the actual proof. Suppose we have two identical objects, with the same (predetermined) properties. We only care about three, initially unknown, properties of the objects, which we can get to know by taking measurements; we'll call them A, B and C. According to our previous assumptions, measuring one property of an object does not affect the other object in any way. We can assign a number to each quality of the object, so that:

$$A = \{0, 1\} \quad B = \{0, 1\} \quad C = \{0, 1\} \quad (12)$$

For example, if the objects are coins and B is their material, $B = 0$ may mean it's made of gold and $B = 1$ may mean it's made of silver. Now, each property has a probability of being found, possibly governed by an unknown parameter λ , which must be between 0 and 1. We can now call $P_{same}(A, B)$ the probability that properties A and B have the same value, which corresponds to cases $A = 1, B = 1$ or $A = 0, B = 0$. Since the objects are identical,

$$P_{same}(A, A) = P_{same}(B, B) = P_{same}(C, C) = 1 \quad (13)$$

Now we can introduce the following inequality, which must always be satisfied within our assumptions:

$$P_{same}(A, B) + P_{same}(A, C) + P_{same}(B, C) \geq 1 \quad (14)$$

This is called a Bell inequality, and it is a central element in all proofs of his theorem. The reasoning behind this inequality is that those probabilities include all the possible outcomes and some of them more than once. Since there are three variables and only two numbers they can take, in each possible configuration, at least two of them must have the same value. By adding those probabilities, every case in which all properties have the same number are counted twice, and every other case once, so the sum of those probabilities must, necessarily, be more than one. But the magic actually happens when you calculate this probability in some quantum-mechanical systems.

To do that, let's consider two qubits in an entangled state:

$$|\Phi^+\rangle = \frac{(|00\rangle + |11\rangle)}{\sqrt{2}} \quad (15)$$

with the two-valued properties A, B and C:

$$A : \begin{cases} |a_0\rangle \equiv |0\rangle \\ |a_1\rangle \equiv |1\rangle \end{cases} \quad B : \begin{cases} |b_0\rangle \equiv \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \\ |b_1\rangle \equiv \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle \end{cases} \quad C : \begin{cases} |c_0\rangle \equiv \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle \\ |c_1\rangle \equiv \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \end{cases} \quad (16)$$

⁶We'll see later that this is but one way to interpret it.

⁷The other conditions and criteria do not really play a key role in his argument.

From these definitions, we see that:

$$|\Phi^+\rangle = \frac{|a_0a_0\rangle + |a_1a_1\rangle}{\sqrt{2}} = \frac{|b_0b_0\rangle + |b_1b_1\rangle}{\sqrt{2}} = \frac{|c_0c_0\rangle + |c_1c_1\rangle}{\sqrt{2}} \quad (17)$$

This manifests the fact that the probability of obtaining the same number in the same properties is equal to unity:

$$P_{same}(A, A) = P_{same}(B, B) = P_{same}(C, C) = 1, \quad (18)$$

in accordance to our system described previously. Now, to calculate the necessary probabilities, we just need to write $|\Phi^+\rangle$ in terms of the eigenstates of the of the properties:

$$|\Phi^+\rangle = \frac{|a_0\rangle(|b_0\rangle + \sqrt{3}|b_1\rangle) + |a_1\rangle(\sqrt{3}|b_0\rangle - |b_1\rangle)}{2\sqrt{2}} \quad (19)$$

The probability of obtaining the same number in A and B will then be the sum of the probabilities of obtaining both zero or both one, calculated with the square modulus of the coefficients of $|a_0\rangle|b_0\rangle$ and $|a_1\rangle|b_1\rangle$, respectively:

$$P_{same}(A, B) = \left|\frac{1}{2\sqrt{2}}\right|^2 + \left|-\frac{1}{2\sqrt{2}}\right|^2 = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \quad (20)$$

Analogously, we can calculate $P_{same}(A, C)$ as follows:

$$|\Phi^+\rangle = \frac{|a_0\rangle(|c_0\rangle + \sqrt{3}|c_1\rangle) - |a_1\rangle(\sqrt{3}|c_0\rangle - |c_1\rangle)}{2\sqrt{2}} \quad (21)$$

$$P_{same}(A, C) = \left|\frac{1}{2\sqrt{2}}\right|^2 + \left|\frac{1}{2\sqrt{2}}\right|^2 = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \quad (22)$$

Finally, we just do the same for $P_{same}(B, C)$:

$$|\Phi^+\rangle = \frac{(|b_0\rangle + \sqrt{3}|b_1\rangle)(|c_0\rangle + \sqrt{3}|c_1\rangle) + (\sqrt{3}|b_0\rangle - |b_1\rangle)(\sqrt{3}|c_0\rangle - |c_1\rangle)}{4\sqrt{2}} \quad (23)$$

$$P_{same}(B, C) = \left|\frac{1 - (\sqrt{3})^2}{4\sqrt{2}}\right|^2 + \left|\frac{(\sqrt{3})^2 - 1}{4\sqrt{2}}\right|^2 = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \quad (24)$$

Plugging the obtained values in (14) yields:

$$\boxed{P_{same}(A, B) + P_{same}(A, C) + P_{same}(B, C) = \frac{3}{4} < 1} \quad (25)$$

And this is our final result: Bell's inequality is violated. This means that one of our assumptions must be **wrong**, and that is, in essence, Bell's theorem.

2.1.2 A more intuitive proof

A proof that relies too much on mathematics is hard to visualize and makes it difficult to get a physical intuition on the topic. That's why, in this section, we will go through a proof based on a phenomenon that is not too hard to find in our everyday life, to later compare it with more sophisticated quantum phenomena.

As stated before, quantum mechanics allows for entangled electrons. But electrons are not the only particle that can become entangled. Many types of qubits⁸ can form an entangled system, one of which is photons. One can produce a pair of entangled photons by a process called **Spontaneous Parametric Down Conversion (SPDC)** [7], by which high-energy photons are directed at a non-linear crystal, like **beta-barium borate (BBO)**, with a low probability of it dividing into two lower-energy entangled photons with the same polarization. Using this, we can construct a similar system to the previous one. In this case, our measurement device will be a polarizing filter [8].

⁸A quantum system initially in a superposition of two possible states, interpreted as 1 or 0, which collapses upon measurement into one of those states (e.g., an electron and its spin, a photon and its polarization, an ion and its energy levels...).

There are different types of polarizers, but we are now only interested in the **linear absorptive** ones. In broad terms, they consist of a film with an internal structure that absorbs vibrational energy in a certain direction. As we know, light is composed of photons, and photons are light waves in the electromagnetic field. These waves oscillate along a concrete direction, maybe even rotating through space, so, classically, we would expect some of their energy to be absorbed when going through a polarizer, leaving only the energy corresponding to the direction of the polariser left. However, this is not the case, and it's because photons have quantized energy. This way, when directed towards a polarizer with a different orientation than the photon, it either goes through or it doesn't. If it gets blocked, its whole energy gets absorbed by the polarizer, and when it passes through, it does so with its whole initial energy, becoming polarized in the direction of the filter. Whether the photon gets blocked or not is a matter of pure probability, determined by that difference in angle between photon polarization and filter orientation. In particular, that probability is of the form (see [9], [10]):

$$P_{through} = \cos^2(\theta), \quad (26)$$

with θ being minimum angle difference between directions of polarization of the polarizer and the photon. This is also in agreement with **Malus' law** [11].

As an example, consider two linear polarizers, A and C, aligned at the same angle and placed one after the other. According to (26), 100% of photons that pass through A should pass through C, as is also verified experimentally. Now, the more we rotate filter C, the lower this percentage becomes, so that when the orientation between the two is perpendicular, no photon that passes through A will be able to pass through C.

For this proof, we will need an entangled pair of photons and three polarizers, oriented in the following way:

$$A \rightarrow 0^\circ \quad B \rightarrow 22.5^\circ \quad C \rightarrow 45^\circ$$

We will then direct each photon towards a filter, with nine possible different setups⁹, depending on our choice. Since our particles are entangled, if one of them makes it through a given filter, this being our **measurement**, the other one will too, and conversely. When choosing different filters, we cannot infer the result of one measurement based on the other. What we can do, however, is assume that the photons have some hidden variable which determines the probability of going through each polarizer ($\checkmark$) or not ($\times$). Given that, we can construct the following inequality:

$$P(A\checkmark, C\times) \leq P(A\checkmark, B\times) + P(B\checkmark, C\times), \quad (27)$$

which can more intuitively be expressed as:

$$P(A\checkmark, B\checkmark, C\times) + P(A\checkmark, B\times, C\times) \leq P(A\checkmark, B\times, C\checkmark) + P(A\checkmark, B\times, C\times) + P(A\times, B\checkmark, C\times) + P(A\checkmark, B\checkmark, C\times)$$

This is equivalent to:

$$0 \leq P(A\checkmark, B\times, C\checkmark) + P(A\times, B\checkmark, C\times),$$

which is obviously true.

We are now ready to check, with our defined settings, whether (27) is satisfied. Considering all the photons that pass through the first filter in each case and using (26), we get:

$$\begin{aligned} P(A\checkmark, C\times) &= 1 - \cos^2(45^\circ - 0^\circ) = 0.5 \\ P(A\checkmark, B\times) &= 1 - \cos^2(22.5^\circ - 0^\circ) \approx 0.1464 \\ P(B\checkmark, C\times) &= 1 - \cos^2(45^\circ - 22.5^\circ) \approx 0.1464 \end{aligned}$$

Lastly:

$$0.5 > 0.1464 + 0.1464 = 0.2928 \quad \Rightarrow \quad \boxed{P(A\checkmark, C\times) > P(A\checkmark, B\times) + P(B\checkmark, C\times)}$$

Therefore, our Bell inequality is violated and we have proven Bell's theorem.

⁹AA, AB, AC, BA, BB, BC, CA, CB, CC.

2.1.3 CHSH inequality

The direct use of probabilities isn't the only way to prove Bell's theorem. The following proof relies on correlations instead of probabilities, and may give another perspective to the topic.

Since we're trying to prove the same theorem, our main initial assumptions are **realism** and **locality**, as usual. Now, let us consider two people: **Alice** and **Bob**. These two people are very far apart (no interaction with each other) and we send a particle to each one of them. Every time we do that, each one of them makes one of two measurements on a normalised¹⁰ property of their particle. As an example, if the properties are the horizontal position of the particle (A for Alice and B for Bob) and the vertical position of the particle (A' for Alice and B' for Bob), the possible results for each measurement are:

$$A = \pm 1 \quad A' = \pm 1 \quad B = \pm 1 \quad B' = \pm 1 \quad (28)$$

Now we can define what we call **CHSH value S**:

$$S = E(A, B) + E(A, B') + E(A' B) - E(A' B'), \quad (29)$$

where $E(X, Y)$ is the **quantum correlation** between measurements. In simple terms, it's the average value of the product between X and Y, which, with our normalised values, is actually the correlation coefficient. If we assume that those values being measured are preexisting and possibly determined by some hidden variable, we could express S_i as:

$$S_i = AB + AB' + A'B - A'B' = (A + A')B + (A - A')B', \quad (30)$$

And then conduct the measuring experiment a lot of times, calculating S_i each time and then taking the average of those values to obtain S. But by expressing S_i as in (30), we can see that, no matter what values our properties take, it is always satisfied that:

$$|S| \leq 2 \quad (31)$$

This expression is known as the **Clauser-Horne-Shimony-Holt (CHSH)** inequality, and it's another Bell inequality, based on correlations or expected values, instead of probabilities [12].

Note that we have not imposed any condition nor correlation on the properties being measured; this is true for any possible predetermined combination of values.

Now, we can analyze what happens in certain quantum systems. Suppose that the particles sent to Alice and Bob are entangled electrons, and each one of them may only measure the spin in one of two perpendicular directions: **a or a'** for Alice and directions **b or b'** for Bob. By normalising the results, the obtained values can only be ± 1 , and we can assign the letters A, A', B, B' to the results of the measurements in the a, a', b and b' directions, respectively. Since the electrons are entangled, measuring the spin of both in the same direction will result in opposite values. The more perpendicular the directions of measurement are, the more uncorrelated the values will be. This is expressed by the following formula:

$$E(X, Y) = -\cos(\theta), \quad (32)$$

with θ being the angle between the directions of measurement of X and Y. This formula can be proven with beyond-our-current-level quantum mechanics and has been proven experimentally many times. To get some intuition behind it, see the following examples:

- $\theta = 0^\circ$: $E(X, Y) = -1$. Same direction, perfect inverse correlation.
- $\theta = 90^\circ$: $E(X, Y) = 0$. Perpendicular directions, no correlation.
- $\theta = 45^\circ$: $E(X, Y) = \frac{\sqrt{2}}{2}$. If $X = +1$, Y may be +1 or -1, with a tendency towards being -1, because the directions of measurement share some component.

In our case, the angles of the directions of measurement will be:

$$a \rightarrow 0^\circ \quad b \rightarrow 45^\circ \quad a' \rightarrow 90^\circ \quad b' \rightarrow -45^\circ$$

Now,

$$E(A, B) = -\cos(45^\circ) = -\frac{\sqrt{2}}{2}$$

¹⁰Normalised meaning that the absolute value of the outcome is defined to be 1.

$$\begin{aligned}
E(A, B') &= -\cos(45^\circ) = -\frac{\sqrt{2}}{2} \\
E(A', B) &= -\cos(45^\circ) = -\frac{\sqrt{2}}{2} \\
E(A', B') &= -\cos(135^\circ) = -\left(-\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2}
\end{aligned}$$

Which yields:

$$S = \left(-\frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right) - \left(+\frac{\sqrt{2}}{2}\right) = -2\sqrt{2}$$

Since, in this case, $|S| > 2$, the CHSH inequality is not satisfied and we infer that one of our hypotheses, either realism or locality (or any other implicit assumption), must be wrong. In fact, the number we obtain corresponds to the maximum discrepancy we can get in a quantum system with regards to the value S, and it is called Tsirelson's bound [13].

2.2 Where the EPR paper went wrong

We have just seen that the EPR paradox has been disproven. That being the case, we can ask the question: "Where did they actually go wrong?"

In essence, the EPR paper was logically correct, as there were no mathematical mistakes. The EPR exposed the idea of the incompatibility between local realism and quantum mechanics, but as they believed on the principle of locality and realism, they said that it was quantum mechanics which was incomplete. However, even if their logic was correct, their assumption before developing the theory was wrong, as the real quantum world does not follow local realism, that only works classically.

Bohr actually responded critically to the paper in the same year, several months later after the publication, attacking some assumptions made as the basis of the previous theory. He started mentioning the criterion of reality, already defined as if the prediction was certain without disturbing the system, the existence of a physical reality is verified. It is true that in this experiment there is no mechanical disturbance, but the action of choosing the measurement of one physical quantity A is, indeed, a condition that influences the system. As an example, if we choose to measure the position of the first particle, then the momentum of the second particle cannot be known because of the lack of physical conditions to define it. To explain this, he defined the **Principle of complementarity** as the mutually exclusive behaviour during measurements of pairs of variables such as the position-momentum pair. Thus, the incompleteness mentioned by the EPR in this case is just a fundamental feature of nature.

Bohr agreed that local realism was incompatible with the quantum physics theory, but also affirming that the wavefunction is the most complete description of a system as there are no hidden variables left to describe, therefore, referring to the possibility of invalidity of the main assumptions made. In his idea, there was no lack of information in the theory, as the limit on prediction is due to the natural behaviour of the systems. This point of view is also called the **Copenhagen interpretation** [14].

Contrary to Bohr, who provided a philosophical argument, in 1964 Bell gave a mathematical response for the EPR supposing true the assumptions made. As known, the results showed the incompatibility of local realism with quantum mechanics, which was a great milestone in quantum mechanics.

2.3 Resulting interpretations of reality

Bell's theorem is a clear example of how we can have rigorous results about something, but still not agree on what they imply, as [15] clearly shows. This is because Bell's theorem does not tell us how reality really works, but rather that certain assumptions about it cannot be true at the same time. We will present here the most important interpretations that result depending on what assumptions we reject.

2.3.1 Copenhagen interpretation

This is actually a group of interpretations based on the premise that **counterfactual definiteness** is not satisfied [16]. That is to reject the idea that all observables have definite values independently of measurements. In that context, if we

measure X, there is not always sense in asking what would have happened had we measured Y instead. It stems from the work of Niels Bohr, Werner Heisenberg, Erwin Schrödinger and others, who were some of the first scientists who set out to develop quantum mechanics, doing so in a probabilistic way, describing particles as in a superposition of states.

It is mostly incompatible with local hidden variable theories, meaning that we cannot know some properties of particles (e.g. spin, momentum, position...) before measurement. This has an obvious problem, which is that there is no description of that transition of states between "before measurement" and "after measurement". This is known as the **measurement problem** [17], and it is the main drawback of the interpretation. However, we don't actually need to know that to describe nature; systems and theories under this interpretation work really well. In **N. David Mermin's** words: *Shut up and calculate!*

2.3.2 Many worlds interpretation

One way to try to solve the measurement problem is to say that each time a measurement is made, reality splits into different worlds, one corresponding to each possible outcome. That is to say that the wave function of a system never collapses, but takes all its possible states in different worlds. Taking **Schrödinger's cat** as an example, before we look into the box, the cat is both dead and alive, but once we do, the world splits into one where the cat is dead and one where the cat is alive, and from the time of measurement on, we'll find ourselves in one of them. This is called **Everett-** or **Many Worlds Interpretation (MWI)** [18].

The wave function collapse that we observe is explained by a process called **quantum decoherence**. This is a way of explaining the apparent wave function collapse. It can be done mathematically [19], but the reasoning behind it is that a measurement involves an interaction of the system with its environment, resulting that in an entanglement between the two. This is such that the environment will behave differently depending on what state the quantum system is in, and depending on the environment that we're part of, we will observe one outcome or the other. However, the other outcome is still there. The initial system is described by a wave function with two possible states, but when it becomes entangled with the environment, it now becomes a wave function including the now bigger system, and this bigger wave function contains the two possibilities, but these two parts become almost orthogonal, so that they cannot interact with each other. That orthogonality corresponds to the fact that they are in different "worlds".

It is not easy to fully grasp the concept with such a short explanation, since it's a very abstract topic. We recommend looking into some citations and doing some research for anyone interested.

2.3.3 Non-local hidden variables

If we choose to reject the assumption that strict locality must hold, we can develop non-local hidden variable theories. The best known one is **Bohmian Mechanics** [20]. In this interpretation, all particles have definite properties at any given moment and are guided by what's known as a **pilot wave**, which in turn depends on the states of all the particles considered. The basis of this theory is to assume that particles can exchange information instantaneously with each other. This explicit non-locality however, does not allow for FTL communication, so it is not in strict disagreement with relativity.

In a quantum system, we still don't know the exact position of particles or other variables. However, it's not because they are in superposition, but because we do not know the underlying mechanism that determines the properties of every particle. In that sense, it is a more intuitive way of looking at quantum mechanics, given that it doesn't rely on superposition.

2.3.4 Superdeterminism

As we said in (2.1.1), **freedom of choice** is an implicit assumption one must make for Bell's theorem to hold. If we choose, however, to accept that those hidden variables are correlated with measurement settings, one can "**escape**" Bell's theorem and develop a local hidden variables theory. This, however, implies that the experimenter does not have complete free will to choose the settings, because everything is absolutely predetermined in the universe, hence the name **superdeterminism** [21].

This hypothesis, like some of the previous ones, is **unfalsifiable**, meaning that it is not possible to prove it wrong nor correct. This argument will always stay as a possibility. Nonetheless, it is not widely accepted, since **free will** is at the base of science and rejecting it would make us have to reconsider all our scientific beliefs.

2.4 Experimental proof

Since the publication of Bell's 1964 paper, numerous experiments have been carried out to prove that Bell's theorem does not only work on paper. The most notable setbacks of this task were the so-called **loopholes**¹¹ [22]. The most important ones are:

- **Detection loophole:** In most experiments, only a small fraction of emitted particles are detected. One could argue that this produces a false violation of the inequality and that this would not happen if we were to detect all of them.
- **Locality loophole:** The two locations of measurement must be sufficiently far apart and the measurements done practically at the same time, so that no signal from one measurement can influence the other. If that is fulfilled, any interaction like "spooky action at a distance" must really be FTL.

Experiments typically use a CHSH-type inequality to tackle the theorem, because they don't require calculating probabilities from the data that is obtained. Furthermore, the particles at play are usually photons, because they are relatively easy to obtain as an entangled pair and also easier to work with.

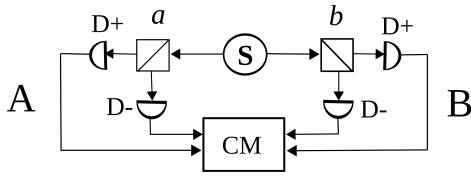


Figure 1: "Two-channel" Bell test scheme [23].

As an example, we have a **CHSH two-channel experiment** [24] (See Fig. 1). We send the entangled photons from S (source) to a and b, which are two-channel polarizers whose orientation can be regulated. The resulting signal gets detected in both sides and counted by the coincidence monitor (CM). In this case, the following formula for S is used¹²:

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b'), \quad (33)$$

where each quantum correlation is calculated as:

$$E = \frac{N_{++} - N_{+-} - N_{-+} + N_{--}}{N_{++} + N_{+-} + N_{-+} + N_{--}} \quad (34)$$

Four subexperiments are conducted to obtain the value of each E , and then the S value is calculated. The angle settings for a, a', b, b' are $0^\circ, 45^\circ, 22.5^\circ$ and 67.5° respectively, as they are the ones that produce the biggest violation of the inequality. As before, if $S > 2$, the inequality is violated.

The most notable experiments are:

- **Kasday, Ullman and Wu (1970):** The first experiment performed, using compton scattering. Did not violate a Bell inequality. [25]
- **Freedman and Clauser (1972):** First experiment to observe a Bell inequality violation, using the Freedman inequality. [26]
- **Weihs et al. (1998):** First experiment to close the locality loophole. This was achieved with a large separation and rapid¹³ choice of measurement settings, which was done using a quantum process to ensure randomness. [29]
- **Rowe et al. (2001):** First experiment to close the detection loophole with two entangled trapped ions, achieving detection efficiencies over 90%. [30]
- **Giustina et al. (2013):** This experiment managed to close the detection loophole for photons for the first time, using highly efficient **transition-edge sensors**. [31]

¹¹These are ways to refute the results of experiments by turning to imperfections or inaccuracies in the fulfillment of formal assumptions.

¹²It's different from (29), but equally valid.

¹³As some experiments did before and as originally proposed by Bell, measurement settings were chosen extremely fast, during the flight of the photons. [27] [28]

- **Hensen et al., Giustina et al., Shalm et al. (2015):** Three significant Bell-tests considered to be **loophole-free**. They managed to simultaneously close the detection, locality and memory¹⁴ loopholes, meaning that any remaining loopholes would be either unfalsifiable or rather convoluted. [32] [33] [34]

All this work led to John Clauser, Alain Aspect and Anton Zeilinger being awarded the 2022 Nobel Prize in Physics for their big contribution in the field [35]. As a curiosity, Alain Aspect led a group that managed to take "photos" of objects using photons that never interacted with the object itself [36].

¹⁴This corresponds to the argument that, since experiments are repeatedly made on the same locations, the measurement devices could have some type of "memory" that makes future outcomes depend on previous ones.

3 Computer simulation to prove Bell's Theorem

3.1 Setup of the simulation. Introduction

Bell's inequalities are now not only a matter of interest for the scientific community, but also for the general public, since in 2022 the Physics Nobel was given to A. Aspect, J. Clauser and A. Zeilinger for their experiments involving **quantum entangled photons**, establishing the violation of Bell's inequality. In this part of the project, we will try to reproduce the quantum experiments that demonstrate this violation, using the open-source quantum computing framework **Qibo**, and MatLab. As it has been already analyzed, there are many Bell-like inequalities. Here we will numerically study two of them, following the scheme proposed in the document attached [37]

The original one, proposed by John Bell in 1964 has already been explained in 2. To briefly summarize that section, that inequality is expressed in terms of 1/2 spin particles. This inequality inspired many others, much more suitable for experimental implementations, such as the so-called ClauserHorneShimonyHolt (CHSH) inequality, proposed in 1969, which will also be analyzed. Lastly, we will study the Bell's-Wigner one, especially convenient due to its intuitive approach, although not being appropriate for practical experiments.

3.2 Bell-Wigner inequality

3.2.1 Theoretical framework

Bell-Wigner inequality is particularly intuitive, and the required mathematics are trivial. That's why the theoretical development will be done here. Let's imagine two particles flying apart, each of them having either a 1/2 spin or a -1/2 spin. Defining three measurement directions, $\vec{a}, \vec{b}, \vec{c}$, it is logical to deduce that, after measuring in one direction from one of the two systems, the measurement in the same direction on the other system will yield the opposite value, since the total spin is $1/2 - 1/2 = 0$. Given so, the only possible results are the following:

Possible states	Particle A	Particle B
N_1	$(\vec{a}+, \vec{b}+, \vec{c}+)$	$(\vec{a}-, \vec{b}-, \vec{c}-)$
N_2	$(\vec{a}+, \vec{b}+, \vec{c}-)$	$(\vec{a}-, \vec{b}-, \vec{c}+)$
N_3	$(\vec{a}+, \vec{b}-, \vec{c}+)$	$(\vec{a}-, \vec{b}+, \vec{c}-)$
N_4	$(\vec{a}+, \vec{b}-, \vec{c}-)$	$(\vec{a}-, \vec{b}+, \vec{c}+)$
N_5	$(\vec{a}-, \vec{b}+, \vec{c}+)$	$(\vec{a}+, \vec{b}-, \vec{c}-)$
N_6	$(\vec{a}-, \vec{b}+, \vec{c}-)$	$(\vec{a}+, \vec{b}-, \vec{c}+)$
N_7	$(\vec{a}-, \vec{b}-, \vec{c}+)$	$(\vec{a}+, \vec{b}+, \vec{c}-)$
N_8	$(\vec{a}-, \vec{b}-, \vec{c}-)$	$(\vec{a}+, \vec{b}+, \vec{c}+)$

Table 1: Possible values of N_i according to Wigner

According to the **hidden variable theory**, since results are fixed before-hand, we can deduce the following expression:

$$P(\vec{a}+, \vec{c}-) \leq P(\vec{a}+, \vec{b}-) + P(\vec{b}+, \vec{c}-) \quad (35)$$

This inequality is the same as the one that has been explained in section (2.1.2) of this study, but applied to this specific scenario. In this particular case, it is equivalent to:

$$N_2 + N_4 \leq N_3 + N_4 + N_2 + N_6 \equiv 0 \leq N_3 + N_6 \quad (36)$$

Which is trivially true. The rigorous proof revolves around noticing that every element in the LHS, is also included inside the RHS of the equation. However, as it will be seen, for certain angles of $\vec{a}, \vec{b}, \vec{c}$, this inequality is false, proving that the **non-deterministic** nature of quantum mechanics can be interpreted as being right. Specifically, that the particle properties are not defined before the measurement, and calculating the probabilities of a certain outcome building on that each state is a subset of N with $i \in \{1, \dots, n\}$ is wrong. More intuitively: measuring in a certain direction doesn't determine the whole system, there can be a variety of different valued correlations.

A trivial deduction is to notice that if we define:

$$Q_W = P(\vec{a}+, \vec{c}-) - P(\vec{a}+, \vec{b}-) - P(\vec{b}+, \vec{c}-) \leq 0 \quad (37)$$

This would hold true according to classical arguments, and false according to quantum mechanics.

3.2.2 Numerical study

Firstly, we will check numerically the last equation. Its is possible to obtain the following expression for $P(\vec{a} \pm +, \vec{b} \pm)$ and $P(\vec{a} \pm +, \vec{b} \mp)$. To see the demonstration, the reader is referred to a more advanced book, covering quantum mechanics with the proper rigour [9]

$$P(\vec{a} \pm +, \vec{b} \pm) = \frac{1}{2} \sin^2 \frac{\theta_{ab}}{2} \quad P(\vec{a} \pm +, \vec{b} \mp) = \frac{1}{2} \cos^2 \frac{\theta_{ab}}{2} \quad (38)$$

Thus, combining both expressions, we can obtain:

$$Q_W = \frac{1}{2} (\sin^2 \frac{\theta_{ab}}{2} - \sin^2 \frac{\theta_{ac}}{2} - \sin^2 \frac{\theta_{cb}}{2}) \quad (39)$$

Plotting this equation yields:

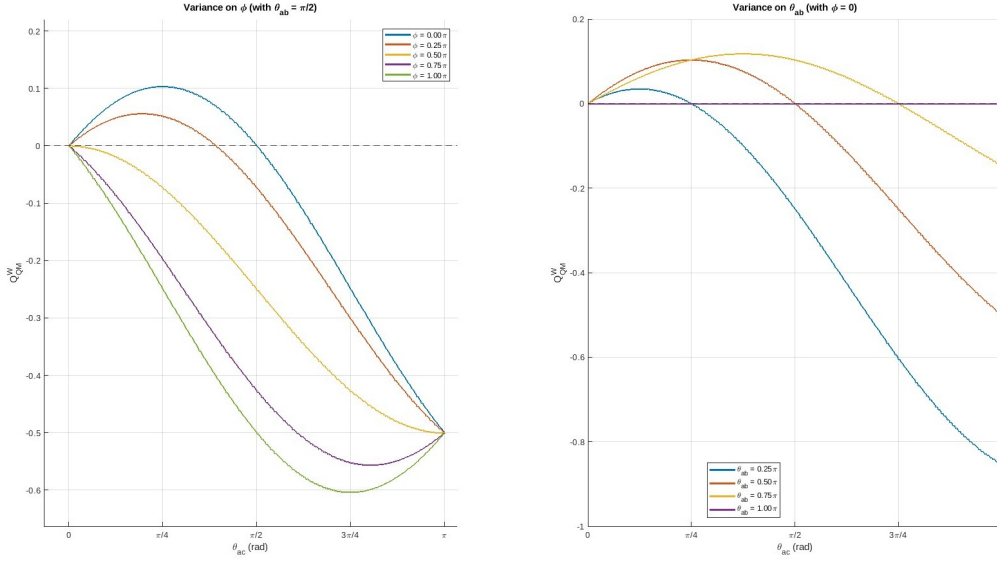


Figure 2: Numerical analysis of Q_W for different systems

Where ϕ is the angle that the $\vec{c}$ forms with the plane defined by $\vec{a}, \vec{b}$. In the left plot, we fixed θ_{ab} and represented Q_W as a function of θ_{ac} for different ϕ values. In the right one, we fixed $\phi = 0$ (meaning that the three axes are coplanar), and plotted Q_W for a range of unequal θ_{ab} . Notice that in both cases, there are certain values for which $Q_W > 0$, meaning that, as we expected, according to quantum mechanics, the inequality doesn't hold true for any given angles. Moreover, this implies that either quantum mechanics or local hidden variables theory are true; their coexistence is impossible. Precisely, what is happening is that, as previously mentioned, quantum states do not carry a label defining its own from: all unmeasured states ($|\Psi^-\rangle$) are indistinguishable from each other.

This analysis has driven us to the conclusion that one of the two theories must be wrong. To prove which one is which, we just have to carry out the experiment: getting some value of $Q_W > 0$ would dismiss Einstein's principle of locality. To make the experiments, we will use the quantum computing framework **Qibo**.

3.2.3 Implementation of the experiment

Before starting to develop any simulation, we will introduce some notions about quantum computing. Firstly, let's suppose that we have a system of n bits. These bits can be either 0 or 1. A quantum bit (qubit) in contrast, is allowed to be in a superposition of states, each of them with a probability associated. For a system with 2 qubits, we can write the general state for the whole system as:

$$|\Psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle \quad (40)$$

Just as a system of orthodox bits, the system of qubits can take 2^n values, however, due to quantum mechanics, each of them has a probability associated, namely $|c_{e_1 \dots e_n}|^2$. The sum of probability is normalized to 1. In our system, we will have two different qubits, and the whole process will be represented as a circuit, as we would do with a system of normal bits. The idea of the simulation is simple: First, we will set up the two particles by entangling them, to end up with particles satisfying the state mentioned in 2.1.1. To imagine how the measurements will be done, we can imagine the usual three orthogonal axes, x, y, z , and overlapping, the three directions a, b, c . Afterward, measurements will be done in the corresponding axes in the following way:

1. $\vec{a}$: is the standard measurement direction. Given so, we align it with the z axis.
2. $\vec{b}$: Since $\vec{b}$ and $\vec{a}$ are coplanar, introducing a gate $R_y(\theta_{ab})$, yields the result along the b axis, rotating an angle θ_{ab} around the y axis
3. To compute the result on c axis, we have to introduce to gates: $R_y(\theta_{ac})$ and $R_z(\phi)$

The setup of the quantum entangled system exposed in also requires some previous concepts. The first one is the quantum gate known as **Hadamard gate**. Hadamard gate is defined mathematically as the matrix that follows:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (41)$$

It is applied to a single qubit. Transforms a given state $|0\rangle, |1\rangle$ into an equally probable superposition of states, while the second qubit remains unaffected:

$$H|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad H|10\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) \quad (42)$$

This Hadamard gate is applied to one of the qubits of the system, namely q_0 . Then, a **CNOT gate** is applied to the other qubit, q_1 , controlled by q_0 . A CNOT gate is a gate that is applied to two qubits. One of them is the **control qubit**, and the other one is the **objective qubit**. Intuitively, this gate inverts the value of the objective qubit given that the control qubit is active. Starting from a pair of electrons in the state $|00\rangle$, then the second qubit will be activated when the output of the Hadamard gate is $|10\rangle$ and will remain inactive otherwise. Given so, the final state of the system will be

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (43)$$

And this is exactly the state previously mentioned as the state of two entangled particles. Intuitively, now we can make measurements over this system, given that if we obtain that one of the two qubits has a certain value, the other one will be completely determined. The code that will be provided represents the circuit that is drawn in the next figure:

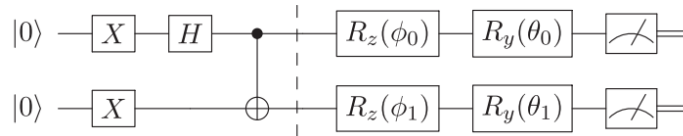


Figure 3: General circuit of the Bell-Wigner inequality experiment [37]

Here is shown how the two qubits are turned to their positive state, the first one goes through a Hadamard gate, then a CNOT gate is applied, and the necessary rotations around the y, z are made (the angles in the figure are an example, in each case we would need to introduce the correct angles to relate the axes that we want to measure $\theta_{ab}, \theta_{cb}, \theta_{ac}, \phi \dots$). In this particular case, the angles will vary in the same way that they did in the previous theoretical plot, in order to compare experiments under the same conditions, and observe whether the inequality is satisfied or not.

Succinctly, this python program encodes the previous circuit, the Bell-Wigner inequality. First, the two bits are set up to their excited value; then, the corresponding gates are applied to create the entangled state. Lastly, the measurements are done in the correct angles. The horizontal axis represents θ_{ac} and the vertical one Q_{QM}^W .

Bell-Wigner inequality implemented in Qibo

```
#Libraries needed to plot the graphs, etc
import numpy as np
import matplotlib.pyplot as plt
from qibo import models, gates, set_backend

set_backend("numpy")

# We initialize a general Bell's inequality circuit
c = models.Circuit(2)

# Both particles are set up to their excited state
c.add(gates.X(0))
c.add(gates.X(1))
# Hadamard gate
c.add(gates.H(0))
# CNOT gate
c.add(gates.CNOT(0, 1))

# Rotations
c.add(gates.RZ(0, -phi_u))
c.add(gates.RY(0, -theta_u))
c.add(gates.RZ(1, -phi_v))
c.add(gates.RY(1, -theta_v))

# Measurements
c.add(gates.M(0, 1))
```

Implementing another code to plot the inequality for certain angle ranges yields:

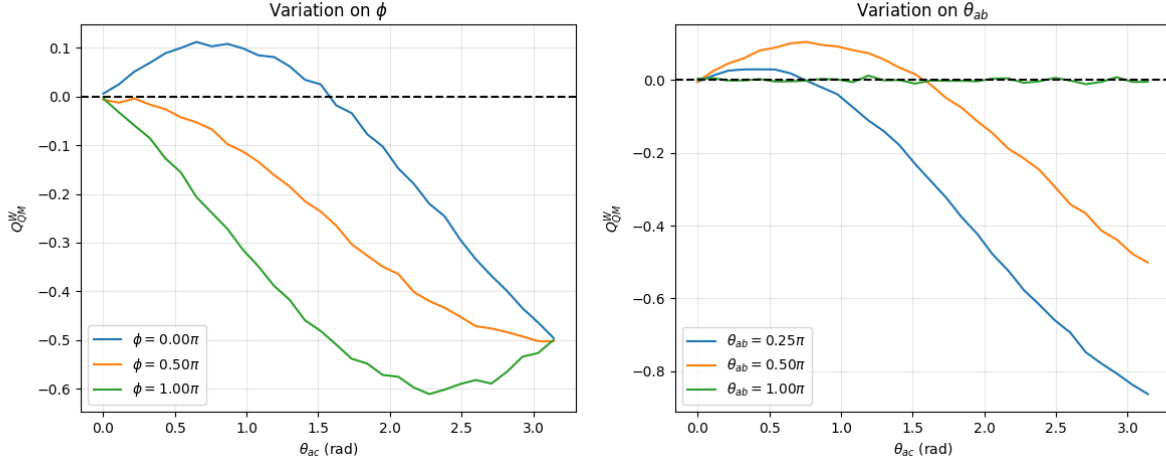


Figure 4: Results of the experiment

Both graphs have been made under the same conditions of the previous plots. Clearly, the curves obtained, reassemble the theoretically predicted results of quantum mechanics, confirming its validity. therefore, hidden variable theory can be disregarded as an incorrect way to interpret reality.

3.3 CHSH inequality

As it has been previously mentioned in section 2.1.3, CHSH inequality is not based on probabilities, much rather on **correlation**. Original Bell's inequality, and the previously explained Bell-Wigner inequality, however, make a specific assumption: **perfect anti-correlation**. Based on the result of measuring an ideal entangled state, a certain configuration of the experiment (for instance $\theta_{ab} = 0$) should always yield opposite results. However, in reality, detectors are not perfect, and therefore assuming they are is not as precise as needed in many experiments. That is why the CHSH inequality doesn't rely on this assumption, since as it was explained in the mentioned section, $|S| \leq 2$ no matter the conditions of the experiment nor the precision of the detectors. Given so, its violation is the most reliable proof of quantum mechanics as a theory. A more detailed text around the derivation and practical implementation of the inequality is attached [38]. Remembering the expression mentioned in 2.1.3, renaming the measure directions as $A = \vec{a}$, $A' = \vec{b}$, $B = \vec{c}$, $B' = \vec{d}$ and transforming the $P()$ notation into $C()$ notation since now we are not dealing with probabilities, but rather correlations, the inequality is transformed into:

$$S = AB - AB' + A'B + A'B' = C(\vec{a}, \vec{c}) - C(\vec{a}, \vec{d}) + C(\vec{b}, \vec{c}) + C(\vec{b}, \vec{d}) \quad (44)$$

Introducing the property explained in 2.1.3, $E(X, Y) = -\cos(\theta_{XY})$, the function to plot transforms (using theoretical quantum mechanic arguments) into:

$$Q_{QM}^S = |\cos(\theta_{ac}) - \cos(\theta_{ad}) + \cos(\theta_{bc}) + \cos(\theta_{bd})| \quad (45)$$

Fixing $\theta_{ab} = \theta_{cd} = \frac{\pi}{2}$ to maximize the violation, constraining $\phi = 0$ (the four axes are coplanar to also maximize violation), and varying θ_{ac} yields the following plot:

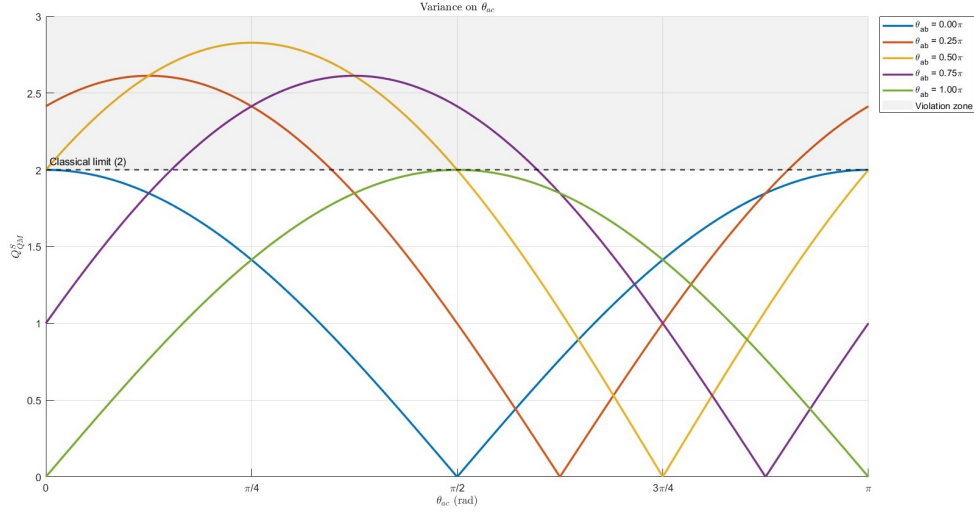


Figure 5: Numerical analysis of Q_{QM}^S for different systems

The following steps are very similar to the ones made with Wigner inequality. Since classical and quantum mechanical arguments can not coexist, we just have to carry out the experiment and observe the behavior of the system. The code implemented in Qibo is very similar to the one for Bell-Wigner inequality, so it will be attached in the appendix. The only difference is that in Bell-Wigner inequality we measure the probability of a single outcome, while in CHSH the main purpose is to measure the **correlation** between the state of the qubits.

The plot yielded by the code is the following:

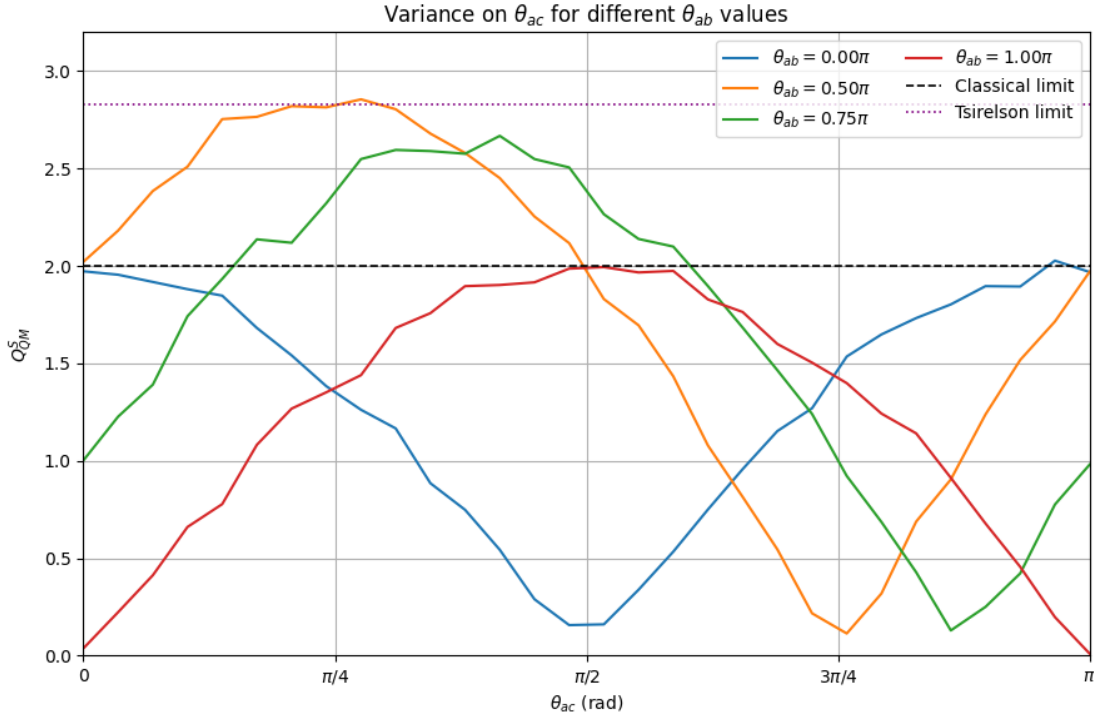


Figure 6: Experimental results on CHSH inequality

We can observe how for certain configurations the classical limit is violated, while the previously mentioned **Tsirelson's limit** is always respected whether we deal with classical or quantum processes. Clearly, the conclusion is the same to the one we got analyzing Bell-Wigner inequality: the experiments confirm the quantum approach to our reality, and the principles defended by Einstein were wrong

3.4 Experimental realism analysis

Since quantum computing is intrinsically probabilistic, the error in the results comes from two primary sources.

3.4.1 Noise error

The first of them is the noise that comes from the precision of our program, in this case MatLab. If we were dealing with real quantum hardware, such as a quantum computer, other possible sources could be:

1. **Thermal noise:** Since the state of the particles is really sensitive to their temperature, any change can affect our system and alter the results
2. **Decoherence:** Any interaction with the environment would result in the loss of quantum properties of the qubits
3. **Detector errors:** There is no such thing as a "perfect device". Given so, laboratory tools make mistakes detecting the presence of particles, etc.
4. **Preparation errors:** Mistakes in the used gates (preparation of the quantum states) could affect the results

Simulations, such as the ones in this study, assume perfect quantum conditions, thereby being very useful to estimate the error in real quantum computing.

3.4.2 Statistical error

Applying well known statistical results, the standard deviation for this experiment is:

$$\sigma = \left(\frac{P(1-P)}{N} \right)^{1/2} \quad (46)$$

Observing the expression allows us to deduce that the error is proportional to $\sigma \approx \frac{1}{\sqrt{N}}$

Thus, to decrease the error in our results, as it is done in any probabilistic process, the solution is to increase the number of times we carry out the experiment.

4 Conclusions

4.1 Conclusions. Analysis of the obtained results

Scientists are always trying to complete, improve and evolve theories, which can define our world. The EPR paper is one of the attempts, and it is important to know that the development of the theory is logically correct, it was only an use of a wrong assumption, specifically the idea of local realism. Each of them individually are viable to define a quantum state, but together is shown incompatible with quantum mechanics. Notwithstanding the fact that the EPR paper followed a false assumption, their theory led to the discovery of entanglement.

When a new theory is born, skepticism is such an essential value for scientists to belong, as a theory is only strong if it can withstand attack and blind acceptance only stops progress. That was exactly what Bohr, Bell and plenty physicists did. In particular, Bohr and Bell attacked the EPR by different ways, Bohr did it philosophically, taking the completeness of the wave function and the Copenhagen interpretation into the debate, and Bell did it mathematically, proving the difference between the EPR theory and reality with numerical proofs.

In our plotting simulations of different Bell inequalities we can visualize the experimental results. We can consider that it shows a significant accuracy comparing with the theoretical estimated results, both respecting the Tsirelson's bound.

Ultimately, the violation of the Bell inequality by the EPR theory is not an unsucces, but a verified knowledge source to better understand quantum mechanics and the universe. This large debate also proves that no matter our believes about the mechanism of the world, the experiments and simulations lead us to the truth.

4.2 Improvement proposals

This study has had many limitations. The main one has been no other than our own knowledge. We would have liked to deepen more in some parts, to be able to explain them in detail and give a more state of the art view on this topic. Nevertheless, we do not believe that this limitations detract from the quality of this study, since as a result of them we have had to seek ways to bypass them, such as using knowledge acquired in other subjects: programming languages (MatLab, Python), Probability and Statistics, and obviously the one we got from Quantum Physics. We have focused into giving a clear and intuitive in-depth explanation of the topic, to familiarize the reader with the subject, followed by a numerical implementation of the experiments. In our opinion, this last part could be expanded analyzing more Bell-like inequalities, and we would also have liked to be able to perform the demonstrations in a more rigorous way.

Another limitation has been the extension of the work. Bell's inequality and entanglement is a very dense topic, with many interesting facts, interpretations and applications. Due to time and extension constraints, we have restricted this work to include only essential information on the topic. We would have loved to include more subject matter within the scope of this study, like *the CHSH game*, deeper explanations of the interpretations of reality and loopholes and the implications that Bell's theorem has had in our modern society, among others.

That said, it was truly a pleasure to make this project.

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5 Appendix

The code to implement CHSH inequality in Qibo is the following:

Implementation of the CHSH inequality in Qibo

```
# Needed libraries
import numpy as np
import matplotlib.pyplot as plt
from qibo import models, gates, set_backend

# The setup of the circuit is the same as the one in Bell-Wigner inequality
c = models.Circuit(2)
c.add(gates.X(0))
c.add(gates.X(1))
c.add(gates.H(0))
c.add(gates.CNOT(0, 1))

c.add(gates.RY(0, -theta_a))
c.add(gates.RY(1, -theta_b))
c.add(gates.M(0, 1))

# Analysis of the results

result = c.execute(nshots=nshots)
freqs = result.frequencies(binary=True)
n = nshots
e = (freqs.get('00', 0) + freqs.get('11', 0) -
     freqs.get('01', 0) - freqs.get('10', 0)) / n
```